Started on	Wednesday, 16 October 2024, 2:34 PM
State	Finished
Completed on	Friday, 8 November 2024, 6:40 PM
Time taken	23 days 4 hours
Marks	20.00/20.00
Grade	**100.00** out of 100.00
Name	DHIRU BHAI 730423104017

Question 1

Correct

Mark 10.00 out of 10.00

Create a class MyTriangle with two static overloaded methods as mentioned below.
a) public static double area(double breadth, double height)
b) public static double area(double a, double b, int angleOfAandB) to compute the area.

The a) method is to compute area by bottom and height:
area = 0.5 * bottom * height

And the b) method is to compute area by two sides a, b and the angle between the two sides (angleOfAandB)
area = 0.5 * a * b * sin(angleOfAandB)

Then create another class Main that contains the main method.
In the main method:
1) Read bottom and height from the Console to compute area by calling the corresponding method you created in MyTriangle
2) Read two sides a, b and angleOfAandB from the Console to compute area by calling the corresponding method you created in MyTriangle.

For example:

Input	Result
5.3 6.2 8.4 7.9 45	16.43 28.23
8.4 7.1 4.8 9.3 60	29.82 -6.80

Answer: (penalty regime: 0 %)

```
import java.text.DecimalFormat;
import java.util.*;
class MyTriangle
{
    public static double area(double a,double b,int c)
    {
        return 0.5*a*b*Math.sin(c);
    }
    public static double area(double a,double b)
    {
        return 0.5*a*b;
    }
}
public class Main
{
    public static void main(String args[])
    {
        DecimalFormat DF = new DecimalFormat("0.00");
        Scanner s = new Scanner(System.in);
        double a = s.nextDouble();
        double b = s.nextDouble();
        double c = s.nextDouble();
        double d = s.nextDouble();
        int f = s.nextInt();
        System.out.println(DF.format(MyTriangle.area(a,b)));
        System.out.println(DF.format(MyTriangle.area(c,d,f)));
    }
}
```

	Input	Expected	Got	
✔	5.3 6.2 8.4 7.9 45	16.43 28.23	16.43 28.23	✔
✔	8.4 7.1 4.8 9.3 60	29.82 -6.80	29.82 -6.80	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 2

Correct

Mark 10.00 out of 10.00

Given a class Cake in the code editor.

a) Based on class Cake and the following table, define two subclasses named as OrderCake and ReadyMadeCake.

	OrderCake	ReadyMadeCake
Additional attribute	weight(kg)	quantity
calcPrice()	rate*weight	rate*quantity

b) By using classes definition from a), write an application program (class Test) that will:

1. Declare an array of 6 cake objects
2. Input data for 3 order cake objects and 3 ready made cake objects and store them into the array
3. Display the total price for all types of cakes
4. Display the information for the cake that has been sold for the highest price.

For example:

Input	Result
ChocoTruffle 400 0.5 Butterscotch 900 1.2 GreenApple 500 0.6 Vanilla 500 3 Blueberry 600 5 Mangodelight 750 6	Total Price for all types of cakes: 10580 Information of the cakes that has been sold for the highest price Vanilla 500.0 Blueberry 600.0 Mangodelight 750.0
Lemonyoghurt 850 2 Redvelvet 950 1.5 Blackforest 800 1 Honeycomb 450 2 Strawberry 500 1 Raspberry 650 2	Total Price for all types of cakes: 6625 Information of the cakes that has been sold for the highest price Lemonyoghurt 850.0 Redvelvet 950.0 Blackforest 800.0

Answer: (penalty regime: 0 %)

Reset answer

```
import java.util.Scanner;
abstract class Cake
{
    protected String name;
    protected double rate;
    public Cake (String n, double r)
    {
        name = n;
        rate = r;
    }
    public abstract double calcPrice();
    public String toString()
    {
        return name + "\t" + rate;
    }
}
class OrderCake extends Cake
{
    protected double weight;
```

```
    public OrderCake(String n, double r, double weight)
    {
        super(n, r);
        this.weight = weight;
    }
    public double calcPrice()
    {
        return rate*weight;
    }
}
class ReadyMadeCake extends Cake
{
    protected int quantity;
    public ReadyMadeCake(String n, double r, int quantity)
    {
        super(n, r);
        this.quantity = quantity;
    }

    public double calcPrice()
    {
        return rate*quantity;
    }
}
public class Test
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        Cake[] info = new Cake[6];
        for(int i=0;i<3;i++)
        {
            info[i] = new OrderCake(sc.next(),sc.nextDouble(),sc.nextDouble(
        }
        for(int i=3;i<6;i++)
        {
            info[i] = new ReadyMadeCake(sc.next(),sc.nextDouble(),sc.nextInt
        }
        int orderprice = 0;
        int readymadeprice = 0;
        for(int i=0;i<3;i++)
        {
            orderprice += info[i].calcPrice();
        }
        for(int i=3;i<6;i++)
        {
            readymadeprice += info[i].calcPrice();
        }
        int totalprice = orderprice + readymadeprice;
        System.out.println("Total Price for all types of cakes: "+totalprice
        System.out.println("Information of the cakes that has been sold for
        if(orderprice > readymadeprice)
        {
            for(int i=0;i<3;i++)
            {
                System.out.println(info[i].name+" "+info[i].rate);
            }
        }
        else
        {
            for(int i=3;i<6;i++)
            {
                System.out.println(info[i].name+" "+info[i].rate);
            }
        }
    }
}
```

	Input	Expected	Got	
✔	ChocoTruffle 400 0.5 Butterscotch 900 1.2 GreenApple 500 0.6 Vanilla 500 3 Blueberry 600 5 Mangodelight 750 6	Total Price for all types of cakes: 10580 Information of the cakes that has been sold for the highest price Vanilla 500.0 Blueberry 600.0 Mangodelight 750.0	Total Price for all types of cakes: 10580 Information of the cakes that has been sold for the highest price Vanilla 500.0 Blueberry 600.0 Mangodelight 750.0	✔
✔	Lemonyoghurt 850 2 Redvelvet 950 1.5 Blackforest 800 1 Honeycomb 450 2 Strawberry 500 1 Raspberry 650 2	Total Price for all types of cakes: 6625 Information of the cakes that has been sold for the highest price Lemonyoghurt 850.0 Redvelvet 950.0 Blackforest 800.0	Total Price for all types of cakes: 6625 Information of the cakes that has been sold for the highest price Lemonyoghurt 850.0 Redvelvet 950.0 Blackforest 800.0	✔

Passed all tests! ✔

▸ **Show/hide question author's solution (Java)**

Correct

Marks for this submission: 10.00/10.00.